Algebra

Independent diagnostic, two proof questions

Time allowed: 45 minutes **Total: 14 marks**

Candidate identifier: _____ **Date:** _____

Instructions. Attempt both questions independently. Show complete arguments, including all relevant cases and any final verification. No internet, calculator, AI tools or hints during the timed sitting. If unfinished, write down any useful progress. Stop the timed work when time is called.

Question 1 **[7 marks]**

Let $f : \mathbb{Z} \to \mathbb{Z}$ satisfy

$$f(x+y) + f(x-y) = 2f(x) + 2y^2$$

for all integers x, y. Find every such function f. Prove both that your list is complete and that every function on your list satisfies the equation.

Question 2 **[7 marks]**

Let a, b, c be positive real numbers. Prove that

$$\frac{a^2}{a+b} + \frac{b^2}{b+c} + \frac{c^2}{c+a} \geq \frac{a+b+c}{2}.$$

Determine exactly when equality holds.

After the sitting (not part of the timed assessment): Record approximate time spent on each question, whether either problem was familiar, and whether you worked beyond the limit. If you need more space, use the separate answer sheets.

Q1: _____ min Q2: _____ min Prior exposure: _____